

PS3: estimating interpreting derivatives and tangent line approx - Answer key qr

1. (Activity 1.5.2) Sentences one

2. (1.5.5.8) coffee cup one

$$(a) F'(10) \approx \frac{F(10.01) - F(9.99)}{0.2} = \frac{141.685 - 141.752}{0.2} = -3.33592$$

(b) deg F per min; (good sentence)

(c) $F'(20) > F'(10)$ bc negative numbers are annoying

(d) neg, inc, c.d., asymptotic to zero

3. (1.8.5.9) Potato

$$(a) F'(30) \approx \frac{F(40) - F(20)}{40 - 20} = \frac{182.4 - 141.7}{20} = 2.035 \text{ deg F per min}$$

$$(b) L(t) = 167.6 + 2.035(t - 30)$$

$$(c) F(33) \approx L(33) = 173.705$$

(d) A bit over bc F is c.d.

4. Approx $2^{5.1}$

- $f'(5) = 2^5 \cdot \ln(2) \approx 22.1807$
- $L(x) = 32 + 22.1807(x - 5)$
- $L(5.1) = 34.21807$
- (Bonus) $L_4(3.89) = 14.78$